

7. A Image with Multiple Channels:

a typical way of handing an image with multiple channels is to simply provide each neuron in a convolutional layer with connections from each channel.

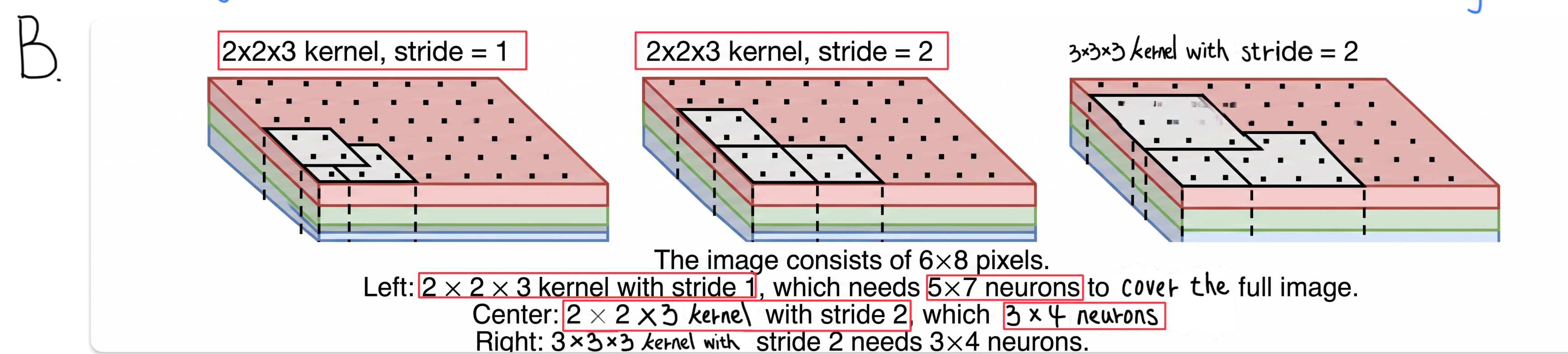
8. Calculating Weights for A Single Neuron in a Convolutional Layer:

$$W_c = (K \times K \times C_{in}) + 1, \left[\begin{array}{l} W_c = \text{num of weights for a single neuron in a convolutional layer.} \\ K \times K \times C_{in} = \text{size of a convolutional kernel.} \\ K = \text{the side length of a convolutional kernel as seen in a 2D plane.} \\ C_{in} = \text{num of input channels in a convolutional layer.} \\ / = \text{num of bias weight for a single neuron in a convolutional layer.} \end{array} \right]$$

9. A Stride:

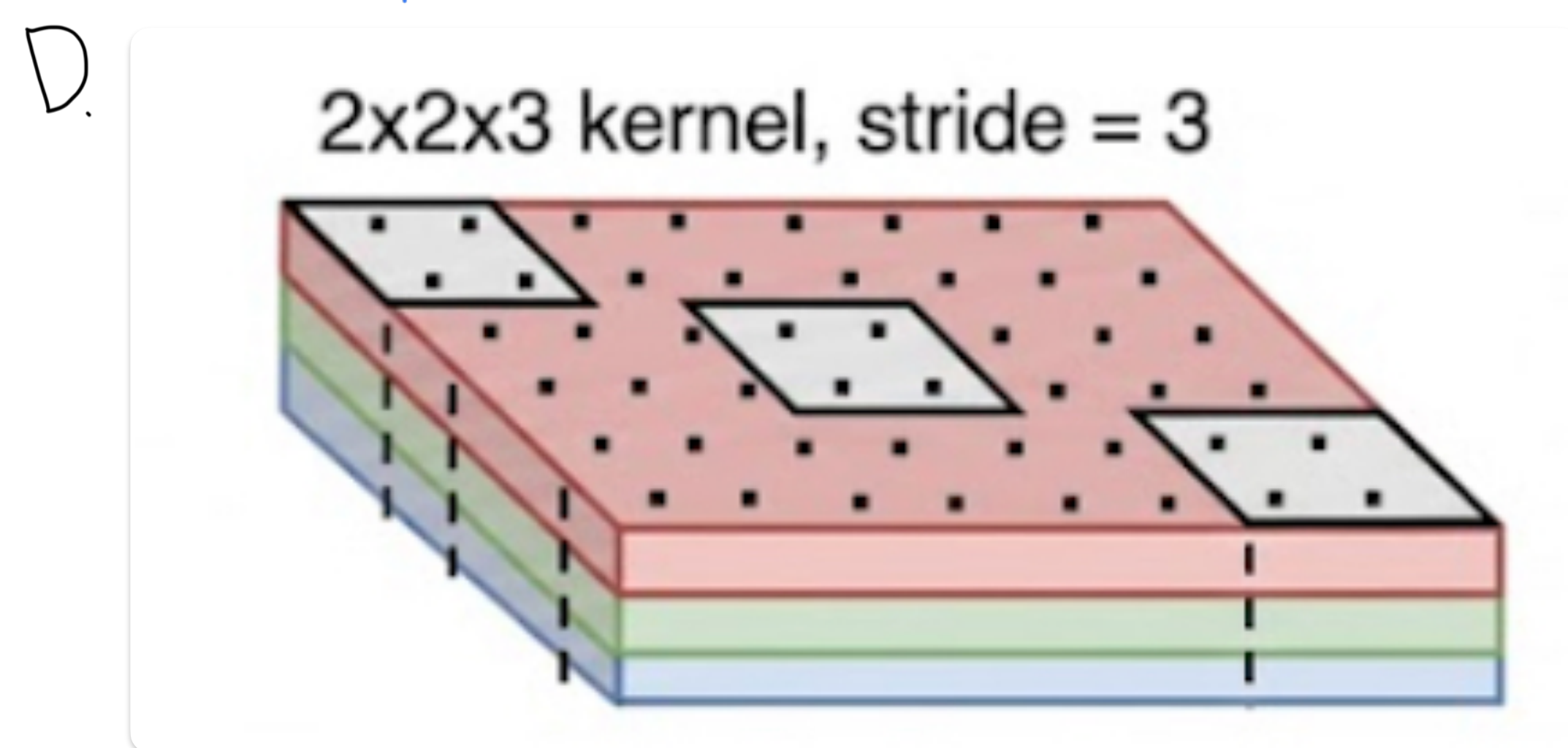
A. a stride represents the interval distance at which the adjacent convolutional kernels slide across an image.

a number of convolutional kernels needed to cover an image is primarily affected by the stride, which the larger the stride, the fewer convolutional kernels are needed to cover an image.



C. a key observation is that "size of a convolutional kernel" and "stride" are orthogonal parameters, but they do interact.

for example, if we choose "2x2x3 kernel" and "stride=3", then some of pixels in the image will not be connected to any convolutional kernel.



10. Padding:

A. padding is an operation that adds extra values which typically are zeros around the borders of an original image or an original feature map before applying convolution kernels to implement filtering processes.

that is, padding effectively enlarges the area of an original image or an original feature map

B. because of a larger image or an larger feature map, we naturally require more neurons which equal to request more convolutional kernels to cover the extra padded area,

this extra amount of neurons exactly maintain the dimension of following feature maps in the next convolutional layer, or without padding, successive convolutions erode the dimension of following feature maps in the next convolutional layer so rapidly to zero, making it impossible to build a deeper CNN.

C. due to a larger image or an larger feature map, the boundary pixels in an original image or the boundary neurons in an original feature map must be allowed to be scanned multiple times by overlapping receptive fields, preventing edge information loss and ensuring edge features are adequately extracted.

D. $P = \frac{K-1}{2}$, $\left[\begin{array}{l} P = \text{num of extra values used to padding.} \\ K = \text{the side length of a convolutional kernel as seen in a 2D plane.} \end{array} \right]$

11. A Rationale for Sparsely Connected Neurons in A Convolutional Layer:

having one neuron process every detail of an entire image is unnecessary and inefficient.

from that perspective, having neurons that simply looks smaller pieces of the image makes sense.

therefore, each neuron doesn't receive data inputs from all pixels in an image. that is, it is not fully connected, but a sparsely connected one.